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Assessment of organic pollution effect considering differences between lotic and lentic stream habitats

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Key words: macroinvertebrates, organic enrichment, riffle-pool habitats, AQEM, stream assessment

Abstract

Based on the requirements of the Water Framework Directive, a macroinvertebrate-based assessment system to evaluate the ecological quality of streams has been developed by AQEM project consortium. In the Czech Republic the impact of organic pollution was principal pressure studied, but some morphological degradation of some sampling sites could not be avoided. A multimetric assessment system for three stream types was developed. Detrended Correspondence Analysis was used for the detection of the response of macroinvertebrate communities to the gradient of organic degradation. Significant relationships between abiotic (BOD, TOC, nutrients) and biotic (saprobic index, ASPT) indicators of organic enrichment/eutrophication were identified. Separate storage of the riffle and pool components of each multi-habitat sample allowed differences between these habitats to be compared in context of the metrics applied in the assessment system. Lotic and lentic habitats differed in taxonomic composition, ecological traits and biotic indices. The separate assessment of the riffle and pool parts of samples provides additional useful information when combined effects of organic pollution and morphological degradation are to be considered.

Introduction

Particulate and dissolved non-living organic matter are important energy inputs in running water ecosystems (Allan, 1995). The natural budget of organic matter in the running waters is frequently supplemented by human activity. Organic enrichment and eutrophication of aquatic ecosystems induced by urbanization and agricultural activities are the oldest and most fully documented forms of pollution. Attention has been paid to this type of pollution from the mid-19th century onwards (Cohn, 1853). Various methods for the evaluation of organic pollution exist, mostly based on lists of indicator organisms. These methods have used plankton, periphyton, macrophytes, fish and benthic macroinvertebrates, but Rosenberg & Resh (1993)

considered macroinvertebrates to represent the most suitable group.

The organic enrichment of streams comprises two aspects: trophism and saprobity. Whereas trophism is the extent and intensity of organic primary production, saprobity is characterized by communities of organisms reflecting the extent and intensity of organic matter decomposition. In practice, the two factors are difficult to separate.

For organic pollution assessment, single metrics such as biotic indices and biotic scores are widely used and are usually based on higher taxonomic levels, (e.g., BMWP, ASPT, Belgian Biotic Index; Metcalfe, 1989). The saprobic system based on benthic macroinvertebrates identified predominantly to the species level, is preferred in Central Europe and is codified

in standards such as the Austrian Standard ÖNORM M 6232, 1997 and the German Standard DIN 38 410 – DEV, 1987, 1992). Based on the traditional saprobic system (Kolkwitz & Marsson, 1902), in the Czech Republic saprobiological methods were further developed by Zelinka & Marvan (1961), Sládeček (1973) and Rotschein (1982). The statutory methods are summarized in the current Czech Standards CSN 75 7716 and CSN 75 7221.

Biotic indices, taxonomic measures and functional feeding group methods are used in ecological assessment; e.g., Rhithron Feeding Type Index – RETI (Schweder, 1990) or the percentages of Ephemeroptera, Plecoptera and Trichoptera taxa. Their suitability for ecological assessment has been tested (Resh et al., 1995; Rawer-Jost et al., 2000).

A multimetric approach of bioassessment frequently used in the USA (Barbour et al., 1992) was also adopted for assessment purposes in the AQEM project (see, e.g., Hering et al., 2004; Dahl et al., 2004; Ofenböck et al., 2004). Multimetric systems based on a set of variables or metrics representing community structure (taxa richness, relative abundance, dominance) pollution tolerance, functional feeding groups, habitat preferences, and life history strategies offer robust and sensitive insights into how an assemblage responds to natural and anthropogenic stressors (Klemm et al., 2003). Multimetric indices have been developed for macroinvertebrates, fish and periphyton (Kerans & Karr, 1994; DeShon, 1995), fish (Miller et al., 1988), periphyton (Hill et al., 2000), and riparian birds (Bryce et al., 2002).

The Water Framework Directive requires assessment of impairment based on comparison with stream type specific conditions. There are several approaches how to deal with fact that parameters used for assessment of human impact naturally fluctuates according to local physical habitat (e.g., hydraulics, substratum, temperature; Statzner & Sperling, 1993) and biogeographic differences depending on geomorphologic and ecoregional conditions (geology, climate and discharge regime; Dolédec et al., 1999). The River Invertebrate Prediction and Classification System (RIVPACS) offer a prediction of the site-specific macroinvertebrate communities to be expected in the absence of major environmental stress (Moss et al., 1999). Development of stream typology is another way how to separate the impairment effect from natural variability in the freshwater ecosystems (see Verdonschot & Nijboer, 2004).

Multivariate approaches, generally based on similarity indices, are also widely used in water quality assessment using macroinvertebrates. The most widely used methods are: TWINSpan (Hill, 1979) and UPGMA (Sneath & Sokal, 1973) for classification and direct and indirect correspondence analysis for gradient analysis (ter Braak & Prentice, 1988). All the methods mentioned above can serve in different ways to assess organic pollution. However, a combination of these methods seems to be powerful tool linking the complexity of macroinvertebrate communities and the requirements of the water management.

Riffle-pool sequences are natural features of running waters associated with erosional and depositional processes (Allan, 1995). Site-specific factors such as local hydraulic conditions and substrate characteristics that influence the macroinvertebrate community structure (Corkum & Ciborowski, 1988) may complicate assessment of impairment. Information needed to compare the capacity of each habitat to indicate the impact of stressors is often limited due to the use of different sampling techniques in riffles and pools. Relatively shallow riffle areas which are easily accessible by wading are studied more frequently than deeper pools (Logan & Brooker, 1983).

In the Czech saprobiological method recently applied in the national monitoring programme, sampling in riffle stretches is preferred, with sampling being carried out in spring in order to reduce effects of habitat heterogeneity and seasonal variation. This approach is supported by partial knowledge of (i) distributional patterns of macroinvertebrates at habitat scale (Logan & Brooker, 1983; Beisel et al., 2000), and (ii) seasonal dynamics in species composition (Furse et al., 1984).

We developed the multimetric assessment system using complex analysis of large dataset. Additionally, the role of site-specific parameters being important for the assessment of water bodies impaired by more than one stressor type was pointed out.

The main objectives of this study are:

- to document the development of a multimetric system for assessing the impact of organic pollution in three different stream types in the Czech Republic;
- to test relationships between environmental parameters related to stressor gradient and benthic community characteristics;
- to analyse the differences in macroinvertebrate communities inhabiting riffle and pool habitats in relation to the potential effect on the developed multimetric systems.

Table 1. Ranges of the environmental characteristics of studied sites

Stream type	C01	C02	C03
Altitude (m a.s.l.)	350–510	200–457	190–633
Catchment area (km ²)	231–1380	9–72	58–460
Distance to source (km ²)	41–94	4–13	18–43
Forest in catchment (%)	17–62	22–100	21–72
Cropland in catchment (%)	6–72	0–67	5–73
Alkalinity (mmol l ⁻¹)	0.2–2.4	2.4–7.2	1.3–5.6

Description of study sites

Since a water body typology is currently being developed in the Czech Republic, the study sites were first divided into types using 'system A' of the Water Framework Directive (WFD; Directive 2000/60/EC – Establishing a Framework for Community Action in the Field of Water Policy). The 37 sites (Fig. 1) are distributed over 12 medium-sized streams in the central sub-alpine mountains (stream type C01), 14 small streams in the Carpathian (C02), and 11 medium-sized streams in the Carpathian (C03). The medium-sized C01 streams had, on average, a larger catchment area, a higher stream order (Strahler) and a lower thalweg slope than C03 streams (Table 1). Geological differences between the regions of C01, C02 and C03 streams are expressed by a higher proportion of flysch and molasse and a lower share of acid silicate rocks in the Carpathian catchments. Consequently, alkalinity and total hardness are higher in flysch and molasse region. Small streams had significantly higher slope of the thalweg than medium-sized rivers ($p < 0.001$). The sites were selected in such way that the largest possible gradient of organic degradation was obtained for each stream type.

Methods

Sampling

Macroinvertebrate samples were taken in spring and summer using the standardized method developed in the AQEM project (Hering et al., 2002, 2004). A multi-habitat scheme was used to sample the habitats proportionally according to their presence within a sampling site. A sample consisted of 20 sampling units, all taken from those microhabitat types at a site that covered at least 5% of the stream section. For each

sampling unit, stationary sampling was carried out by positioning a long handled net (500 µm mesh size) upright on the stream bed and disturbing the upstream substratum over an area of 0.25×0.25 m (0.0625 m²). Material collected from all microhabitats was pooled, though the lotic and lentic part of the sample was kept separate. Samples were preserved immediately in 4% formaldehyde. All macroinvertebrates were sorted, identified and counted. Before statistical analyses, biological data were taxonomically adjusted using a standardization procedure described by Schmidt-Kloiber (2003). The entire AQEM methodology is described in detail in the 'Manual for application of the AQEM system' available at www.aqem.de (AQEM consortium, 2002).

Environmental parameters

A set of environmental parameters was collected for each site. Hydromorphological, hydrological, land cover and other features at habitat, site, stretch (floodplain) and catchment level are described in detail in the AQEM method manual (AQEM consortium, 2002). BOD (biochemical oxygen demand), TOC (total organic carbon) and concentration of nutrients were recorded as the water chemistry parameters potentially related to organic pollution and/or eutrophication. All GIS analyses were based on stream network and catchments data provided by the Water Research Institute TGM in Prague and by Water Research Institute in Bratislava. Land cover data were obtained from the CORINE system (provided by the Ministry of Environment of the Czech Republic) using ARCview 3.1 software. The digital atlas of the Czech Republic ('Atlas GEOCR 500') produced by the Czech Geological Institute was used as the source of information about the catchment geology. As a measure of microhabitat heterogeneity the Shannon–Wiener diversity index, evenness and microhabitat richness were used. Formulae usually used for species data were modified by the substitution of species quantity with the proportion of microhabitats.

Biotic metrics

Karr & Chu (1999) defined metrics as measurable parts or processes of biological system empirically shown to change in value along a gradient of human influence. During the AQEM project, two tools for calculation of biota-based metrics were developed. ATIC is a collection of macros developed in Microsoft Excel which was used for evaluation of sampling programme

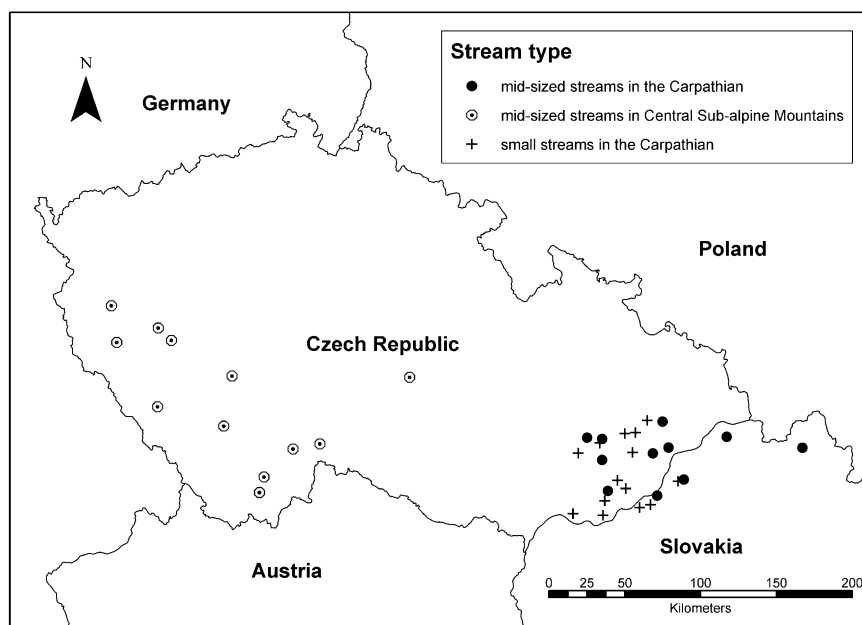


Figure 1. Map of study area showing position of sampling sites (37) representing the three stream types.

results. Each country generated a type-specific list of metrics, class boundaries and multimetric formulae. This information was incorporated into the assessment software facilitated the calculation of metrics and the classification of samples into the ecological quality classes (in terms of WFD).

The Czech saprobic system contains 940 macroinvertebrate taxa. Dominant taxonomic group are Diptera with 192 scoring taxa (78 chironomid taxa) followed by Trichoptera (130), Coleoptera (121), Plecoptera (101), Oligochaeta (96), Ephemeroptera (78) and Mollusca (63). The calculation of the formula is described in the AQEM method manual (AQEM consortium, 2002) and is identical to the Austrian one. The Czech and Austrian saprobic indices differ in saprobic valences and in individual saprobic indices for individual taxa. In the Czech system, taxa with individual saprobic index between 1 and 2 (53%) dominate while only 12 taxa (1%) reach values between 3 and 4. Chironomidae and Oligochaeta are the most frequently scored taxa groups in polysaprobic valence within evaluated dataset (55% of total scoring taxa).

Statistical analysis

Detrended Correspondence Analysis (DCA), using 2nd order polynomials, biplot scaling and inter-sample distances, was undertaken on macroinvertebrate abundance data. For this purpose, data were

$\log(x + 1)$ transformed and rare taxa were down-weighted. The relationships between the structure of macroinvertebrate communities and the values of environmental parameters were further analysed by a nonparametric correlation (Spearman rank correlation coefficient).

As a basic tool for macroinvertebrate community classification, TWINSpan (Two Way Indicator Species Analysis) was used (Hill, 1979). Pseudospecies were defined by three cut levels of 0, 10 and 100 individuals. No weighting or omissions were applied to the dataset. Data of each stream type were analysed separately.

The samples were also classified into groups using hierarchical agglomerative clustering method UPGMA (unweighted pair groups with arithmetic averaging) based on Bray–Curtis dissimilarity index calculated as a distance measure by a ‘macro’ developed for Statistica for Windows version 6.0. This statistical package was also used for cluster analyses, One-way ANOVA was applied for testing differences in biotic and abiotic parameters.

The Kruskal–Wallis test was used for pre-selection of variables with statistically significant differences between classification groups.

Development of multimetric assessment systems

System development consists from the following steps: testing of community response to stressor gradient and divisive clustering based on community structure (a), transformation of TWINSpan groups into groups being differentiated by saprobic index values (b), setting provisional class boundaries for saprobic index (c), selection of other core metrics (d), and calculation of the stressor-specific quality class (e).

(a) The abiotic classification, based on parameters indicating organic pollution, was tested against the biotic metrics but gave unsatisfactory results. Since the project consortium had a common target to construct multimetric assessment systems by integrating information from a range of biotic metrics, the initial steps were to classify samples and to select metrics with the highest discriminatory power. Application of several clustering methods showed that TWINSpan provided results related to known indicators of organic pollution. Data from individual stream types were processed separately.

(b) Using the theoretical range of the saprobic index scores, groups were merged into 4 or 5 groups ('bad' status conditions for stream types C01 and C03 were not identified in the dataset). As a guide for the fusion of groups, a non-significant difference between groups or, in the case of groups with low number of sites, a large overlap of saprobic index values were used. The position in the hierarchical structure of the dendrogram was also taken into account. Differences in saprobic index values between merged groups were statistically significant (Kruskal–Wallis test, $p < 0.01$ for all stream types).

(c) Draft class boundaries were set for the saprobic system generated using box-plots characteristics of merged TWINSpan groups and by estimation when an inter-class gap occurred without values. External data set from reference sites were used for setting the class boundary between high and good status. These draft class boundaries were used for draft classification of samples based on saprobic index values. This was necessary because of the modification of multivariate results for efficient using in multimetric systems. In this way, 'draft classification' of samples was subsequently used for selection of another metrics forming multimetric system together with saprobic index.

(d) All biotic and abiotic parameters were tested with the Kruskal–Wallis test for differences between the 'draft groups'. The aim was also that the system

should incorporate representatives of all biotic metric types (sensitive taxa indices, diversity, ecological traits). Another criterion for metric selection was a relationship to stressor gradient and differentiation among groups. Selected metrics were tested by stepwise discriminant analysis to find the best combination of metrics that provided the highest percentage of correctly classified samples (compared to 'draft groups' membership). Discriminant analysis was used with caution as a supportive tool since the data did not fully meet the criteria for parametric analysis. The discriminatory power of metrics for the differentiation of individual classes was tested by non-parametric post-hoc tests. The final set of core metrics included in the multimetric system and their class boundaries used in the multimetric formula are listed in Table 3.

(e) For each sample, each metric value was classified into a metric-specific class (score) and in case of an overlap of class ranges the mean values of both classes were used. These metric results were subsequently averaged and rounded into a final class value for the respective sample - the stressor-specific quality class.

Results

Relationships of macroinvertebrate variability to environmental gradients

Preliminary, top-down stream typology was confirmed because all three stream types formed fairly well separated groups of sites in the DCA ordination diagram (spring and summer data were analysed separately). Separation of sites within stream types was most marked on the second axis; these sample scores were significantly correlated ($p < 0.001$) with geological characteristics of the catchments, differentiating the flysch and molasse part of the Carpathian ecoregion and the acid silicate rocks of the Central Sub-alpine Mountains ecoregion (Spearman $R = 0.55$ and $R = -0.63$, respectively). Statistically significant correlations with distance to source, catchment area, and Strahler stream order reflected the differences between the macroinvertebrate communities that are linked to stream size. Stream types were better separated by spring data than summer data, where an overlap of the medium-sized streams of both ecoregions was evident.

The first axis of DCA, which explained 14% (summer data) and 15% (spring data) of macroinvertebrate variability, was significantly correlated with the known environmental indicators of organic enrichment (BOD, TOC, nutrient concentration). This

indicated that site selection had reached its main goal – to include a gradient of organic pollution.

Riffle proportion, biochemical oxygen demand (BOD), proportion of arable land in the catchment, mean slope of the thalweg, and chloride, nitrate, and total phosphate concentrations had comparable relationships to the first axis scores in both seasons (all Spearman R with $p < 0.001$). Percentage of organic mud, current velocity, and total organic carbon (TOC) explained macroinvertebrate variability better in spring than in summer. In contrast, water depth and coverage of shoreline with woody riparian vegetation were more important in summer. Amongst those parameters crucial for an explanation of biota variability, there are factors operating in various spatial scales from microhabitat to catchment level (Table 2).

The final classification in values of the Czech saprobic index, with class boundaries is shown in Fig. 2. The difference between the original TWINSPAN classification, sorted by the saprobic index, and the final classification illustrates the relationships between multivariate system based on taxonomic structure of entire communities and scoring system based on sensitive indicators. About half of all the 74 samples remained in the same class in both systems. The vast majority of shifts involved changes of only one class, and only 2 samples differed in two classes between the applied classifications.

Core metrics

For the purpose of the core metrics selection, the data from individual stream types were analysed separately. The Czech saprobic index, ASPT (Average Score per Taxon) and the RETI index (Rhithron Feeding Type Index) were chosen for medium-sized streams of stream type C01. Differences among the TWINSPAN groups and the final classifications were confirmed by a Kruskal–Wallis ANOVA at level $p < 0.001$ (with the exception of ASPT and the TWINSPAN classification, $p < 0.01$). Besides the Czech saprobic index, two additional metrics were selected for the assessment system of small streams in the Carpathian (C02) – the number of stonefly taxa and the number of mayfly taxa. Both classifications produced groups significantly different in terms of core metrics at level $p < 0.01$. Furthermore, for the saprobic index and the number of stonefly taxa, the final classification increased statistical significance to $p < 0.001$. It was relatively difficult to find metrics which, together with saprobic index, would form a reliable multi-

metric assessment system for stream type C03. The number of EPT taxa (Ephemeroptera + Plecoptera + Trichoptera) was found to give the best fit. Results of a non-parametric ANOVA for final classification illustrated the discriminatory power of this parameter ($p < 0.001$), exceeding the saprobic index ($p < 0.01$). Number of EPT taxa was linearly correlated with the number of EPT individuals in stream types C01 and C02 ($R^2 = 0.29$, $p < 0.01$ and $R^2 = 0.42$, $p < 0.001$).

A common element of the multimetric assessment systems developed for the three stream types studied in the Czech Republic is the saprobic index (Czech version) used as tolerance metric representing community characteristic based on sensitive taxa. This is a traditional metric used in assessment systems focused on the evaluation of impairment by stressors related to organic pollution or eutrophication.

The Czech saprobic index differed significantly among the stream types due to lower values in small streams (Kruskal–Wallis ANOVA, $N = 74$, $p < 0.05$). The difference between the median saprobic index values of between small (C02) and medium-sized (C01 and C03) streams is equal to 0.5 (C01 = 1.97; C02 = 1.45; C03 = 1.99), but only the difference between stream types C02 and C03 is statistically significant ($p < 0.05$). Statistically significant differences related to stream size were also recorded by the RETI index – its values were higher in small streams ($p < 0.001$). The remaining core metrics did not differ significantly among stream types.

Since the dataset contained relatively few samples in each individual season, the common assessment criteria (core metrics, class boundaries) were applied to spring and summer samples together. Final class differences between seasons were higher in medium-sized streams than in small streams.

Linking biotic metrics to environmental parameters

The final classification was significantly related to some other biotic metrics (Fig. 3) that were excluded from the multimetric system in order to keep the balance between the number of samples used for development of the assessment system and the number of metrics used for classification. These parameters offer a potential for future improvements of class discrimination when a greater number of samples are available.

The responses of the individual metrics to water chemistry, habitat characteristics and catchment

Table 2. Correlation between the first DCA-axis and the selected environmental parameters. Spearman correlation coefficients for spring and summer samples were evaluated separately, $N = 37$

Parameter	Spring		Summer	
	<i>R</i>	<i>p</i> -level	<i>R</i>	<i>p</i> -level
% of organic mud	0.60	<0.001	0.41	<0.05
Chloride	0.86	<0.001	0.79	<0.001
BOD 5	0.78	<0.001	0.44	<0.01
Ammonium	0.59	<0.001	0.38	<0.05
Nitrite	0.69	<0.001	0.73	<0.001
Nitrate	0.62	<0.001	0.38	<0.05
Total phosphate	0.71	<0.001	0.74	<0.001
Mean slope of the thalweg	−0.53	<0.001	−0.64	<0.001
TOC	0.59	<0.001	0.41	<0.05
% of cropland in catchment	0.73	<0.001	0.68	<0.001
Percent of riffle	−0.70	<0.001	−0.70	<0.001
Altitude	−0.44	<0.01	−0.29	N.S.
Conductivity	0.39	<0.05	0.16	N.S.
Mean depth	0.34	<0.05	0.57	<0.001
Mean current velocity	−0.41	<0.05	−0.21	N.S.
Maximum current velocity	−0.38	<0.05	−0.18	N.S.
% of shoreline covered with woody riparian vegetation ^a	−0.41	<0.05	−0.48	<0.01
Maximum depth	0.09	N.S.	0.37	<0.05
Stream density	−0.30	N.S.	−0.42	<0.05

^aAverage of both banks.

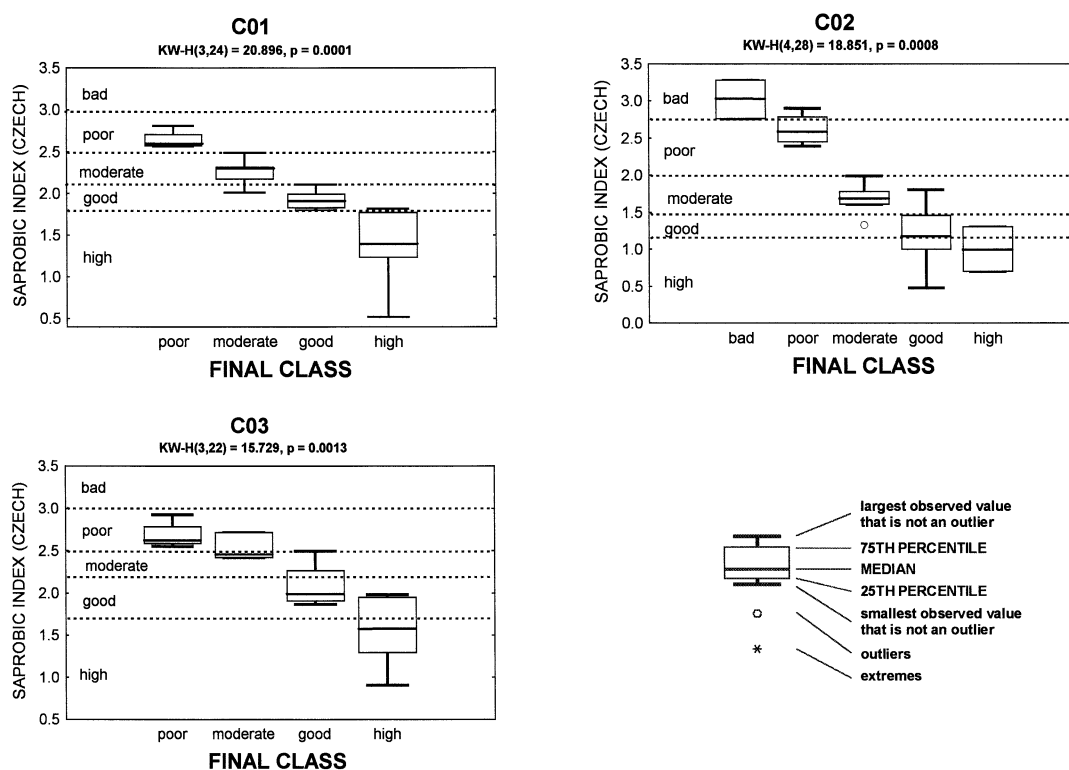


Figure 2. Final multimetric classification displayed as a box-whiskers plots of the Czech saprobic index with its class boundaries represented by dotted lines; results of the Kruskal–Wallis ANOVA reflect difference among classes.

Table 3. Core metrics and their class boundaries for individual ecological status classes

Stream type	Core metrics	Ecological status classes				
		5 (High)	4 (Good)	3 (Moderate)	2 (Poor)	1 (Bad)
C01	Saprobic index (CZ)	≤ 1.80	1.81–2.10	2.11–2.50	2.51–3.00	$\geq 3.01^a$
	ASPT	≥ 6.01	≥ 6.01	5.21–6.00	4.01–5.20	$\leq 4.00^a$
	RETI	≥ 0.471	0.281–0.470	0.151–0.280	0.051–0.150	$\leq 0.050^a$
C02	Saprobic index (CZ)	≤ 1.20	1.21–1.50	1.51–2.00	2.01–2.70	≥ 2.71
	Plecoptera (number of taxa)	≥ 5	2–4	1	0	0
	Ephemeroptera (number of taxa)	≥ 12	7–11	4–7	≤ 3	≤ 3
C03	Saprobic index (CZ)	≤ 1.70	1.71–2.20	2.21–2.50	2.51–3.00	$\geq 3.01^a$
	EPT (number of taxa)	≥ 18	15–18	8–15	5–8	$\leq 5^a$

^aIndicates class boundary set up without sufficient data set.

characteristics were analysed. In the medium-sized streams, the organic enrichment indicated both by BOD and TOC was positively correlated with Czech saprobic index (Spearman R , $p < 0.001$) and negatively with ASPT ($p < 0.001$) and RETI ($p < 0.001$). On the other hand, in the small streams, these biotic metrics were much more highly correlated to nutrient gradients (nitrite, nitrate, phosphate) and to the proportion of arable land in the catchment. The RETI index, for example, was negatively correlated to nitrite, ortho-phosphate concentrations and also to the proportion of cropland in the catchment (Spearman R , $p < 0.001$).

The biotic metrics were also related to microhabitat characteristics. For example, IBE versus proportion of coarse particulate organic matter – CPOM ($R = 0.49$, $p < 0.001$), and Czech saprobic index values versus the proportion of organic mud ($R = 0.53$, $p < 0.001$) in medium-sized streams. In small streams, there were distinct relationships between proportion of CPOM and ASPT or RETI (both $R = -0.58$, $p < 0.001$). Coverage of fine sediments (argyllal + FPOM + organic mud) at the sites (C02 type) was related to the proportion of cropland in the catchment (Spearman R , $p < 0.01$). Habitat preference of benthic fauna to fine sediments (POM, pelal, argyllal) and gathering collectors density exhibited relationships to reach-scale characteristics in the medium-sized rivers (share of the fine substrata cover) while the percent of cropland in catchment was more important in the small streams.

Many presumable relationships between site characteristics and trophic strategies or microhabitat preferences of macroinvertebrate communities have been detected. For instance, pelal (mud with grain size < 0.063 mm) microhabitat preferences of macroinver-

tebrates were related to the percentage of cropland in catchment, percentage of passive filter feeders was related to the proportion of megalithal (grain size > 40 cm) in the stream substratum and proportion of grazers in the macroinvertebrate community was related to proportion of riffles at the site (all Spearman R significant at $p < 0.001$).

Differences between lotic and lentic habitats

The ecological attributes of organic pollution pre-determines possible differences between lotic (riffles) and lentic (pools) stream habitats. Effects of organic matter degradation in streams are significantly influenced by local conditions of current velocity, substratum and channel morphology. The proportion of both habitats varied within the studied sites. In each season, only lentic habitats were present at 4 sites, whilst only lotic habitats were present at 9 sites. However, a mixture of both habitats occurred at 19 sites. At the remaining 5 sites only lotic habitats were present in spring but these differentiated into a mixture of both habitats in summer.

Despite the high variability of depth and current velocity, statistically significant differences between habitats were observed ($p < 0.001$). Current velocity decreased from riffles to pools but also from spring to summer. Considerable decline in current in summer, associated with seasonal flow regime, is most evident in small streams where mean current velocity decreased from 0.54 m s^{-1} to 0.19 m s^{-1} in riffles, and from 0.23 m s^{-1} to 0.03 m s^{-1} in pools (Table 4). Lower mean current velocity in C01 than C03 could be due to a lower slope of C01 streams.

Differences in microhabitat diversity were tested by calculating the Shannon–Wiener index of diversity,

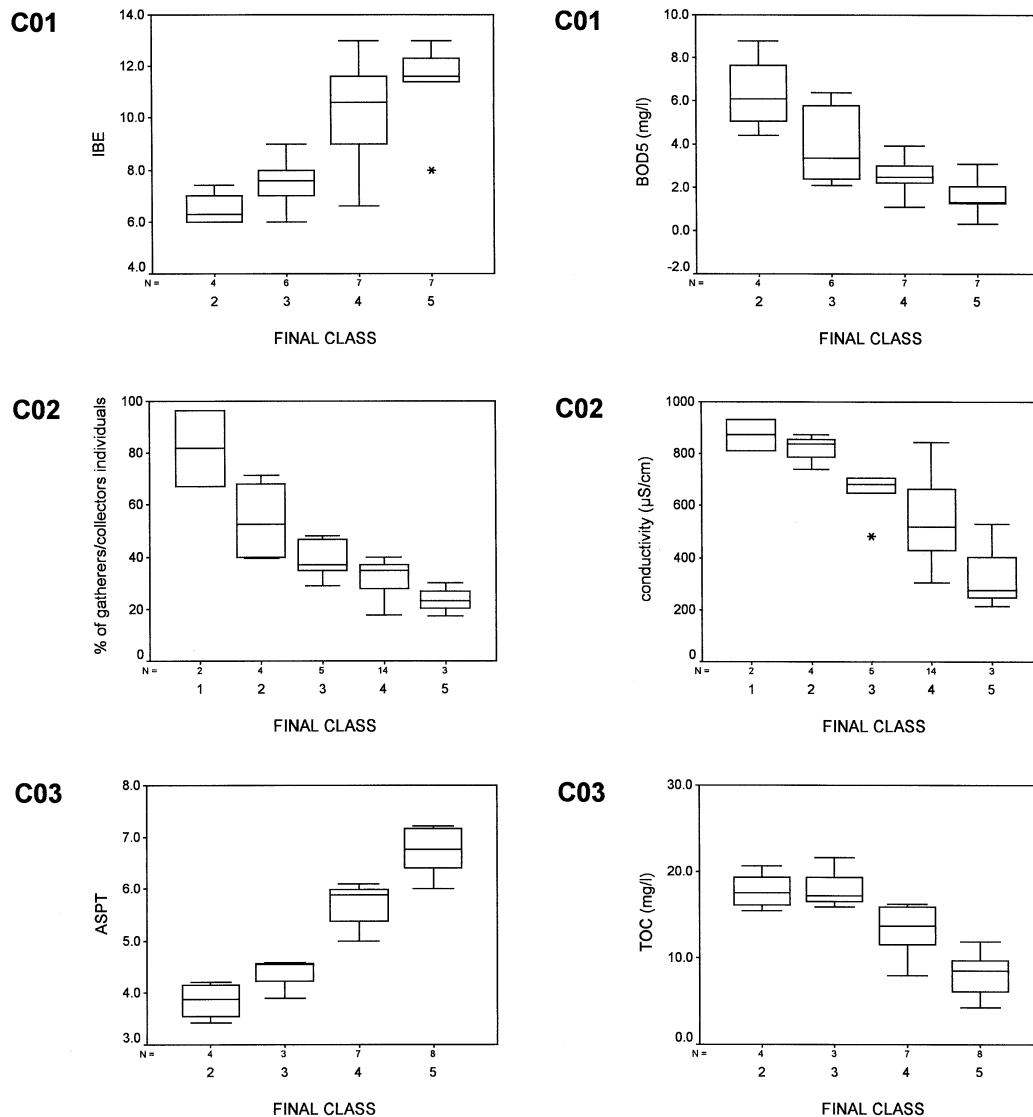


Figure 3. Selected metrics and environmental parameters showing significant discriminatory power for class separation in individual stream types, that were not used in the final assessment system (Kruskal–Wallis test $p < 0.01$). Legend: see Figure 2; final class 1 = bad status, class 5 = high status.

the evenness and the microhabitat richness, using percentage data of each microhabitat. Differences between habitats were registered only in the C02 type with the diversity index and the number of microhabitats both higher in riffles than pools ($p < 0.05$). Although lower microhabitat heterogeneity would also be expected in the pools of C01 and C03 stream types, due to sampling only taking place in wadable stretches, microhabitat diversity was comparable in both habitats. However, microhabitat composition differs significantly between habitats, with a higher pro-

portion of organic mud, psammal/psammopelal and akal in pools, and a higher proportion of macrolithal (20–40 cm) and mesolithal (6–20 cm) in riffles.

The subset of samples collected from sites where both habitats were present was analysed for differences in biotic parameters of macroinvertebrate communities. In total 43 samples included riffle and pool components (8 samples of C01 type, 22 samples of C02 type and 13 samples of C03 type). Besides differences in habitat preference, feeding strategies, and current preference, significant differences in the val-

Table 4. Current velocity (m s^{-1}) summarized for individual stream types, seasons and habitats (mean $\pm$ standard error)

Stream type	Season	Riffle	Pool
C01	Spring	0.57 ± 0.030	0.24 ± 0.015
C01	Summer	0.42 ± 0.028	0.04 ± 0.009
C02	Spring	0.54 ± 0.025	0.23 ± 0.021
C02	Summer	0.19 ± 0.01	0.03 ± 0.006
C03	Spring	0.74 ± 0.023	0.29 ± 0.024
C03	Summer	0.37 ± 0.021	0.13 ± 0.019

ues of the core metrics were also found (Table 5). These results should be considered with caution that values of some metrics (especially the taxa richness metrics) can be related to the sampled area which was in some cases different for riffle and pool habitats. Riffles are characterized by higher proportion of fauna preferring lithal, rheophils, passive filtrators, grazers/scrapers and total abundance of macroinvertebrates. Taxa with pelal and particulate organic matter microhabitat preference, active filtrators and gatherers/collectors exhibited higher densities in pools. A higher Czech saprobic index and RETI in pools and a higher number of EPT taxa in riffles caused 14 samples taken from riffle part of a site to be classified as in a better class than the related pool (8 pairs of samples differed by 1 class and 6 by two classes) samples. Most of the samples were classified in the same class for both habitats (23) and in 6 samples the pools were calculated to be one class better than the riffles. These results are important consideration in the context of an assessment system whose range of application includes sites with varying mixture of habitats.

Another subset of 16 samples with an equal proportion of riffle and pools was analysed to see whether the taxonomic composition of the samples was similar in both habitats. Bray–Curtis dissimilarity index values between habitats were negatively correlated with the Czech saprobic index of mixed samples ($R^2 = 0.39$, $p < 0.01$). There was no significant difference ($N = 16$, $p > 0.05$) found in total number of taxa, number of Ephemeroptera taxa, number of Plecoptera taxa, number of families and total density (N m^{-2}) between the two habitats. Number of EPT taxa and percentage of EPT individuals were higher in riffle habitat (Wilcoxon test, $N = 16$, $p < 0.001$). Slightly

Table 5. Biotic metrics with statistically significant difference between lotic and lentic habitats; Wilcoxon test for dependent design were used ($N = 43$)

Metrics	<i>p</i> -level	Higher median
Habitat preference – lithal [%]	<0.001	Riffle
Current preferences – rheophils [%]	<0.001	Riffle
Passive filter feeders [%]	<0.001	Riffle
Habitat preference – pelal [%]	<0.001	Pool
Coleoptera [number of taxa]	<0.001	Riffle
Grazer and scrapers [%]	<0.001	Riffle
Czech saprobic index	<0.001	Pool
Active filter feeders [%]	<0.001	Pool
Diptera [%]	<0.001	Pool
EPT [%]	<0.001	Riffle
Habitat preference – POM [%]	<0.001	Pool
IBE	<0.001	Riffle
Trichoptera [%]	<0.001	Riffle
Gatherers/collectors [%]	<0.001	Pool
Trichoptera [number of taxa]	<0.001	Riffle
RETI	<0.001	Riffle
EPT [number of taxa]	<0.01	Riffle
Ephemeroptera [%]	<0.01	Riffle
Number of taxa	<0.05	Riffle
Abundance [ind. m^{-2}]	<0.05	Riffle

higher RETI index in riffles was on the border of statistical significance (Sign test was non significant and Wilcoxon test was significant at $p < 0.05$ level).

Discussion

Development of multimetric assessment system

The multimetric system for the assessment of organic pollution and eutrophication based on macroinvertebrates was developed for three stream types in the Czech Republic. The wide application of the multimetric concept in the U.S.A. shown that it is an effective tool for the assessment of a broad range of human impacts on aquatic ecosystems (Barbour et al., 1995). This type of assessment system is based on the deviation from the type-specific reference condition. Stream typologies developed in individual European countries differ in parameters being used (biotic/abiotic) and also in general approach (top-down/bottom-up).

Detrended correspondence analysis of data collected in the Czech Republic showed that community

structure was related to gradients of parameters used in typology (geology in the catchment, stream size). Separation of the geological types on the second axis of correspondence analysis was also found by Charvet et al. (2000) in semi-natural streams in France.

Core metrics were selected to be predominantly based on the detailed taxonomic identification of macroinvertebrates, different types of metrics to be involved and to maintain metrics traditionally used in the region. An example of the application of the last rule is the saprobic system. Although the saprobic system was criticized for not working in turbulent streams (Chandler, 1970), and for not taking into account of local factors (Hynes, 1960; Statzner & Sperling, 1993), traditional usage in some countries of Central Europe initiated its adaptation to the Water Framework Directive requirements (Rolauffs et al., 2004). An assumption that saprobic index is a plausible measure of the response of the macroinvertebrate community to organic pollution reflected in the observed data and the Czech version of the saprobic index has been incorporated in all three stream-type specific assessment systems. This index was significantly correlated to BOD, TOC and organic mud coverage at the site. Metcalfe-Smith (1994) referred limitations for using biotic indices as bioassessment tools. Scoring within stream type specific conditions and long term records demonstrating relationships to the environmental indicators of organic pollution partially reduced these limits. The only biotic index operating at family level which exhibited a fit with the assessment system developed for stream type C01 was ASPT (Average Score Per Taxa). This index combines pollution tolerance (mostly organic) and a diversity measure (De Pauw et al., 1992).

Richness measures are frequently included in multimetric systems (Roy et al., 2003; Klemm et al., 2003). Perhaps due to taxonomic difficulties with the typical Chironomidae and Oligochaeta fauna in the lentic benthos the taxa richness is more frequently used for assessment of disturbance effects in streams and rivers than in stagnant waters (Resh & McElvay, 1993). Resh et al. (2000) found that richness metrics were most accurate in detecting impairment and avoiding classification errors. As the richness metrics most supplemented saprobic system information were found EPT, Ephemeroptera and Plecoptera taxa richness. Linear relationship between density and taxa richness of EPT taxa in stream types C01 and C02 indicated importance of precise following the sampling manual to collect a comparable samples. This results

should be taken into consideration in case of application of subsampling procedure or other manipulation with the sample size.

Breaking the EPT into separate orders, as we used for stream type C02, may allow more effective diagnosis of the stressor among impaired waters because each order responds differently to a different stressor (Clements, 1994; Fore et al., 1996; Klemm et al., 2003). Study at unimpacted sites had shown that EPT taxa richness was more stable and more predictable than total taxa richness (Lenat & Penrose, 1996).

Since the functional relationships among species in terms of ecosystem processing and uniqueness of species are unpredictable (Covich, 1996) it is useful to combine biodiversity measures with metrics based on species traits. In the background of developed assessment systems are statistically significant relationships between feeding and habitat preferences of macroinvertebrate communities and impairment characteristics. However case studies focused on microhabitat-scale pattern would be needed for a better understanding of the interrelationships between natural patterns and effect of impairment. It is not clear how well some changes in these ecological traits measures are related to changes in water or habitat quality (Barbour et al., 1999). Furthermore, metrics based on functional feeding groups can be highly variable depending on both instar and food conditions (Cranston, 1990; Hildrew et al., 1985). The effect of organic pollution on the feeding strategies of macroinvertebrates is expressed by the RETI index. This index represents the percentage of grazers/scrapers, shredders and xylophagous taxa.

Considering number of sites involved in the study we would recommend further validation and tuning of the system. Development of systems for additional stream types and different types of stressors would be needed. Resh et al. (2000) compared multimetric approaches with multivariate models and they recommended combination of both methods to enhance water quality monitoring and assessment programs. The Czech Republic has good preconditions to develop an assessment system based on the database of reference sites (PERLA project unpublished data) and the stressor-specific multimetric systems in future.

Relationships among biotic and environmental measures of impairment

Relationships between macroinvertebrate community structure and stressor intensity (organic pollution)

were detected at level of individual taxa composition (DCA ordination) and also at derived metrics. First DCA axis was significantly correlated to abiotic parameters indicating organic pollution/eutrophication at various spatial scales. It is in agreement with results of Lammert & Allan (1999) who found that detection of significant relationships between biotic integrity of streams and local or catchment factors are results of processes that occur at multiple scales. Roy et al. (2003), studying effect of urbanisation, found strong relationships between catchment land cover and stream biota, but they also reported that reach-scale variables had higher capabilities for predicting variation in biotic indices than land cover parameters. The relatively low proportion of community structure variability explained by DCA ordination axis (first axis 15%) might be related to the character of multihabitat sample in which macroinvertebrates taken from all microhabitats with at least 5% cover of sampled stretch are pooled.

Oxygen deficiency represents the primary abiotic indicator of the intensity of organic matter decomposition. Biochemical oxygen demand (BOD) is a measure of microbial respiration and it is frequently used as an indicator of organic pollution. Saprobic index, ASPT and RETI indices were significantly correlated with BOD and these relationships were particularly evident in medium-sized rivers. The fact that individual taxa differ in sensitivity to low oxygen saturation levels is used for the organic enrichment assessment. The number of Plecoptera taxa, used as a core metric for the stream type C02, decreases with an increase in stressor pressure. Absence of Plecoptera in poor and bad class sites showed that the relationship between a metric and a stressor gradient need not be necessarily be linear within the full extent of stressor intensity. Generally, Plecoptera provide a high bioindication potential (especially where oxygen-demand pollutants are concerned) for the assessment of small streams because of their sensitivity to the concentration of dissolved oxygen (e.g., Baumann, 1979).

The input of organic matter into channels modifies the bottom substrate characteristics. Sedimented particulate organic matter forms patches dependent on the instream hydraulic conditions. Because fine organic particles have hydrological properties similar to those of silt (Inman, 1949), food quantity is rarely a limiting factor in silt-bottom habitats (Ward, 1992). These reach-scale geomorphic and chemical parameters are related to catchment land cover (Roy et al., 2003). This was in agreement with our results from small streams

where coverage of fine sediments at the sites was significantly related to proportion of cropland in the catchment. Feeding strategies and microhabitat preferences are key biotic aspects indicating these kinds of degradation. We found predictable relationships between microhabitats proportions and the microhabitat preference of benthic fauna (fine sediments) which indicated suitability of developed methodology for studying this type of linkages. Charvet et al. (2000) pointed the potential of various biological and ecological traits for detection of different human impacts. Dolédec et al. (1999) found that biological traits were less confounded by natural spatial gradients and indicated human impact better than ecological traits. In this study feeding strategies did not differ significantly between medium-sized rivers from Central highlands and the Carpathians, but relative abundance of shredders was significantly higher in the small streams than in both medium-sized rivers. Microhabitat characteristics exhibited were related to the feeding strategies (fine sediments vs. gathering collectors) and to biotic indices (CPOM vs. IBE; organic mud vs. Czech saprobic index). Fine sediments suspension and deposition affects benthic macroinvertebrates by altering substrate composition, changing suitability of the substrate for some taxa, increasing drift, affecting respiration and affecting feeding activities (Wood & Armitage, 1997).

Since the key products of organic matter decomposition are nutrients, eutrophication is very closely linked to organic pollution. Addition of nitrate and phosphate in combination with a greater light input results in enhanced plant growth. In these conditions, the primary effects of eutrophication on benthic macroinvertebrates are shifts in the feeding strategy of communities towards herbivory. A negative correlation between percentage shoreline covered by woody riparian vegetation and the DCA axis I representing fauna response to stressor supported studies indicating the role of riparian buffers in protecting water quality (e.g., Gregory et al., 1991). On the other hand, autochthonous primary production is an important source of organic matter and its decomposition may result in phenomena comparable with the effect of allochthonous organic enrichment (increased BOD, amount of particulate organic matter, low dissolved oxygen concentration). Furthermore, decomposition of organic matter in streams may be influenced by the concentration of nutrients (Meyer & Johnson, 1983).

Effects of organic pollution and eutrophication on stream benthic fauna are linked to each other by pro-

cesses of organic matter and nutrients transformations. Common assessment system for both stressors is recommended and the effect of each of them can be indicated by individual stressor specific metrics. Future multi-scale investigation of physical and biotic structure and function of stream ecosystems would be needed for the assessment of impairment and also for the development of restoration strategies.

Differences between riffle and pool macroinvertebrate fauna

Spatial and temporal variability complicate detection of stressor effects, and natural gradients must be considered in sampling or evaluation design (Lammert & Allan, 1999). Factors related to organic pollution or eutrophication (dissolved oxygen content, substrate character and nutrient cycling) differ between lotic and lentic habitats. The flow type (turbulent/laminar) affects the water aeration, supplying the water with oxygen and enhancing decomposition. Seasonal and habitat differences in discharge and hydraulic conditions were documented at studied streams. Microhabitat diversity differed significantly between habitats only in small streams, but composition differences were significant also for medium-sized rivers. As shown by the analysis of between-habitats taxonomical similarity related to saprobic index, organic pollution reduces the difference between biota of naturally different units of a stream ecosystem. This could be also explained by a higher environmental heterogeneity in slightly impacted sites that resulted in a more diverse macroinvertebrate community. Logan & Brooker (1983) found that riffles and pools at the same site were generally more closely related (cluster analyses of Jaccard similarity coefficient) to each other than to other riffles and pools. As we tried to cluster samples based on Bray–Curtis similarity (unpublished results) we obtained more inter-site similarity than the above-mentioned study which used data from several investigations. Riffle-pool sequences are also important for nutrient exchange between a channel and a hyporheic zone in terms of nutrient spiraling concept (Newbold et al., 1982). Anthropogenic input of organic matter (sewage and domestic waste) and nutrients (fertilizers) alters the energy base and trophic pathways of communities (Covich et al., 1999).

Statistically significant differences of environmental parameters and biotic metrics, including some core metrics, showed that habitat-specific parameters may enhance separation of stressor effect from over-

all variability. Total density was significantly higher in riffles compared with pools and this is in agreement with conclusion reported by Logan & Brooker (1983). Scarsbrook & Townsend (1993) found higher taxa richness in riffles than pools in a stable stream but no difference between the two habitats in an unstable stream. Logan & Brooker (1983) found no significant differences in richness at several taxonomic levels between habitats. In our study no significant difference was found in total taxa richness, number of families, number of Ephemeroptera taxa and number of Plecoptera taxa between riffle and pool habitats of 16 samples having equal proportion of both habitats. Number of EPT taxa and percentage of EPT individuals were higher in riffle habitat. A review of biodiversity variation at various spatial scales by Vinson & Hawkins (1998) also contains several examples showing significant interhabitat variation in richness. Differences were found also in Czech Saprobic index, IBE, relative abundance of gatherers/collectors, grazers, active filter feeders which had direct effect on the final class of ecological quality.

Although only limited part of dataset was suitable for evaluation of habitat difference we suggest for future studies to test using separate metrics and/or scoring of metrics for riffle and pool habitats. Otherwise the final results would be influenced by the proportion of habitats within sampled stretch. This is associated with the selection of a ‘representative stretch’ has limited repeatability. Klemm et al. (2003) separated riffle and pool components of a sample to develop a system for assessing a particular reach on only the dominant habitat of that reach. They found that the Macroinvertebrate Biotic Integrity Index, with its development based mainly on riffle-dominated sites, had lower applicability in pool-dominated reaches. Logan & Brooker (1983) reported that effects of particulate solids were greater on the pool fauna than the riffle fauna. On the other hand, macroinvertebrate assessment model for small-sized rivers in the southern Italian Apennines was based on data from depositional areas only (Buffagni et al., 2004). Harper et al. (1993) determined functional habitats as useful approach for river conservation bridging the gap between biological and geomorphological surveys. Our results showed that separate assessment of the riffle and pool parts of samples provides additional information useful when combined effects of organic pollution and morphological degradation are to be considered.

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References

- Allan, J. D., 1995. Stream Ecology – Structure and Function of Running Waters. Chapman & Hall, London, 388 pp.
- AQEM consortium, 2002. Manual for the application of the AQEM method. A comprehensive method to assess European streams using benthic macroinvertebrates, developed for the purpose of the Water Framework Directive. Version 1.0, February 2002.
- Barbour, M. T., J. L. Plafkin, B. P. Bradley, C. G. Graves & R. W. Wiseman, 1992. Evaluation of EPA's Rapid Bioassessment benthic metrics: metric redundancy and variability among reference stream types. *Environ. Toxicol. Chem.* 11: 437–449.
- Barbour, M. T., J. B. Stribling & J. R. Karr, 1995. The multimetric approach for establishing biocriteria and measuring biological condition. In Davis, W. S. & T. P. Simon (eds), *Biological Assessment and Criteria. Tools for Water Resources Planning and Decision Making*. Lewis Publishers, Boca Raton, Florida: 63–77.
- Barbour, M. T., J. Gerritsen, B. D. Snyder & J. B. Stribling, 1999. Rapid bioassessment protocols for use in streams and wadable rivers: periphyton, benthic macroinvertebrates and fish, second edition. EPA 841-B-99-002. Office of Water, US Environmental Protection Agency, Washington DC.
- Baumann, R. W., 1979. Nearctic stonefly genera as indicators of ecological parameters (Plecoptera: Insecta). *Great Basin Nat.* 39: 241–244.
- Beisel, J. N., P. Usseglio-Polatera & J. C. Moreteau, 2000. The spatial heterogeneity of a river bottom: a key factor determining macroinvertebrate communities. *Hydrobiologia* 422/423: 163–171.
- Bryce, S. A., R. M. Hughes & P. R. Kaufmann, 2002. Development of a bird integrity index: using bird assemblages as indicators of riparian condition. *Environ. Manag.* 30: 294–310.
- Buffagni, A., S. Erba, M. Cazzola & J. L. Kemp, 2004. The AQEM multimetric system for the southern Italian Apennines: assessing the impact of water quality and habitat degradation on pool macroinvertebrates in Mediterranean rivers. *Hydrobiologia* 516: 313–329.
- Chandler, J. R., 1970. A biological approach to water quality management. *Water Pollut. Control* 69: 415–421.
- Charvet, S., B. Statzner, P. Usseglio-Polatera & B. Dumont, 2000. Traits of benthic macroinvertebrates in semi-natural French streams: an initial application to biomonitoring in Europe. *Freshwat. Biol.* 43: 277–296.
- Clements, W. H., 1994. Benthic invertebrate community response to heavy metals in the Upper Arkansas River Basin, Colorado. *J. North Am. Benthol. Soc.* 13: 30–44.
- Cohn, F., 1853. Über lebende Organismen im Trinkwasser. *Günsberg's Z. Klin. Med.* 4: 229–237.
- Corkum, L. D. & J. J. H. Ciborowski, 1988. Use of alternative classification in studying broad-scale distributional patterns of lotic invertebrates. *J. North Am. Benthol. Soc.* 7: 167–179.
- Covich, A. P., 1996. Stream biodiversity and ecosystem processes. *Bull. North Am. Benthol. Soc.* 13: 294–303.
- Covich, A. P., M. A. Palmer & T. A. Crowl, 1999. The role of benthic invertebrate species in freshwater ecosystems: zoobenthic species influence energy flows and nutrient cycling. *BioScience* 49: 119–127.
- Cranston, P. S., 1990. Biomonitoring and invertebrate taxonomy. *Environ. Monit. Assessment* 14: 265–273.
- CSN 75 7221, 1998. Water Quality – Classification of Surface Water Quality. Czech Technical State Standard, Czech Standards Institute, Prague, 10 pp.
- CSN 75 7716, 1998. Water Quality, Biological Analysis, Determination of Saprobic Index. Czech Technical State Standard, Czech Standards Institute, Prague, 174 pp.
- Dahl, J., R. K. Johnson & L. Sandin, 2004. Detection of organic pollution of streams in southern Sweden using benthic macroinvertebrates. *Hydrobiologia* 516: 161–172.
- De Pauw, N. P., P. F. Guetti, D. P. Manzini & R. Spaggiari, 1992. Biological assessment methods for running waters. In Newman, P., A. Piavaux & R. Sweeting (eds), *River Water Quality: Ecological Assessment and Control*. EC, Brussels: 217–248.
- DeShon, J. E., 1995. Development and applicability of the invertebrate community index. In Davis W.S. & T.P. Simon (eds), *Biological Assessment and Criteria: Tools for Water Resource Planning and Decision-Making*. Lewis Publishers, Boca Raton, Florida.
- DEV (Deutsches Institut für Normung e.V.), 1987. Biologisch-ökologische Gewässeruntersuchung (Gruppe M): Allgemeine Hinweise, Planung und Durchführung von Fließgewässeruntersuchungen (M1). In *Deutsche Einheitsverfahren zur Wasser, Abwasser und Schlammuntersuchung*. VCH Verlagsgesellschaft mbH, Weinheim, 13 pp.
- DEV (Deutsches Institut für Normung e.V.), 1992. Biologisch-ökologische Gewässeruntersuchung: Bestimmung des Saprobienindex (M2). In *Deutsche Einheitsverfahren zur Wasser, Abwasser und Schlammuntersuchung*. VCH Verlagsgesellschaft mbH, Weinheim, 13 pp.
- Dolédéc, S., B. Statzner & M. Bournard, 1999. Species traits for future biomonitoring across ecoregions: patterns along a human-impacted river. *Freshwat. Biol.* 42: 737–758.
- Fore, L. S., J. R. Karr, & R. W. Wiseman, 1996. Assessing invertebrate responses to human activities: evaluating alternative approaches. *J. North Am. Benthol. Soc.* 15: 212–231.
- Furse, M. T., D. Moss, J. F. Wright & P. D. Armitage, 1984. The influence of seasonal and taxonomic factors on the ordination and classification of running-water sites in Great Britain and on the prediction of their macro-invertebrate communities. *Freshwat. Biol.* 14: 257–280.
- Gregory, S. V., F. J. Swanson, W. A. McKee & K. W. Cummins, 1991. An ecosystem perspective of riparian zones. *Bioscience* 41: 540–551.
- Harper D. M., C. D. Smith & P. J. Barham, 1993. Habitats as the building blocks for river conservation assessment. In Boon, P. J., P. Callow & G. E. Petts (eds), *River Conservation and Management*. Wiley & Sons, New York: 311–319.
- Hering, D., A. Buffagni, O. Moog, L. Sandin, M. Sommerhäuser, I. Stubauer, C. Feld, R.K. Johnson, P. Pinto, N. Skoulikidis, P. Verdonschot & S. Zahradkova, 2002. The development of a system to assess the ecological quality of streams based on macroinver-

- tebrates – design of the sampling programme within the AQEM project. *Int. Rev. Hydrobiol.*
- Hering, D., O. Moog, L. Sandin & P. F. M. Verdonschot, 2004. Overview and application of the AQEM assessment system. *Hydrobiologia* 516: 1–20.
- Hildrew, A. G., C. R. Townsend & A. Hasham, 1985. The predatory Chironomidae of iron-rich stream: feeding ecology and food web structure. *Ecol. Entomol.* 10: 403–414.
- Hill, M. O., 1979. TWINSPLAN – A FORTRAN program for arranging multivariate data in an ordered two-way table by classification of individuals and attributes. Cornell University Ithaca, N.Y., 90 pp.
- Hill, B. H., A. T. Herlihy, P. R. Kaufmann, R. J. Stevenson, F. H. McCormick & C. B. Johnson, 2000. Use of periphyton assemblage data as an index of biotic integrity. *J. North Am. Benthol. Soc.* 19: 50–67.
- Hynes, H. B. N., 1960. *The Biology of Polluted Waters*. Liverpool University Press, Liverpool.
- Inman, D. L., 1949. Sorting of sediments in the light of fluid mechanics. *J. Sediment Petrol.* 19: 51–70.
- Karr, J. R. & E. W. Chu 1999. *Restoring Life in Running Waters: Better Biological Monitoring*. Island Press, Washington D.C.
- Kerans, B. L. & J. R. Karr, 1994. A benthic index of biotic integrity (B-IBI) for rivers of the Tennessee Valley. *Ecol. Appl.* 4: 768–785.
- Klemm, J. D., K. A. Blocksom, F. A. Fulk, A. T. Herlihy, R. M. Hughes, P. R. Kaufmann, D. V. Peck, J. L. Stoddard, W. T. Thoeny, M. B. Griffith & W. S. Davis, 2003. Development and evaluation of a macroinvertebrate biotic integrity index (MBII) for regionally assessing mid-atlantic highlands streams. *Environ. Manag.* 31: 656–669.
- Kolkwitz, R. & M. Marsson, 1902. Grundsätze für die biologische Beurteilung des Wassers nach seiner Flora und Fauna. *Mitt. aus d. Kgl. Prüfungsanstalt für Wasserversorgung u. Abwasserbeseitigung*, Berlin 1: 33–72.
- Lammert, M. & J. D. Allan, 1999. Assessing biotic integrity of streams: Effects of scale in measuring the influence of land use/cover and habitat structure on fish and macroinvertebrates. *Environ. Manag.* 23: 257–270.
- Lenat, D. R. & D. L. Penrose, 1996. History of the EPT taxa richness metric. *Bull. North Am. Benthol. Soc.* 13: 305–307.
- Logan P. & M. P. Brooker, 1983. The macroinvertebrate faunas of riffles and pools. *Wat. Res.* 17: 263–270.
- Metcalfe, J. L., 1989. Biological water quality assessment of running waters based on macroinvertebrate communities: history and present status in Europe. *Envir. Pollut.* 60: 101–139.
- Metcalfe-Smith, J. L., 1994. Biological water quality assessment of rivers: use of macroinvertebrate communities. In Calow, P. & G. E. Petts (eds), *The Rivers Handbook, Hydrological and Ecological Principles*, Volume 2. Blackwell Science, Oxford: 144–170.
- Meyer, J. L. & C. Johnson, 1983. The influence of elevated nitrate concentration on rate of leaf decomposition in a stream. *Freshwat. Biol.* 13: 177–183.
- Miller, D. L., P. M. Leonard, R. M. Hughes, J. R. Karr, P. R. Moyle, L. H. Schrader, B. A. Thompson, R. A. Daniels, K. D. Fausch, G. A. Fitzhugh, J. R. Gammon, D. B., Halliwell, P. L. Angermeier & D. J. Orth, 1988. Regional applications of an index of biotic integrity for use in water resource management. *Fisheries* 13: 12–20.
- Moss, D., M. T. Furse, J. F. Wright, P. D. Armitage, 1987. The prediction of the macroinvertebrate fauna of unpolluted running-water sites in Great Britain using environmental data. *Freshwat. Biol.* 17: 41–52.
- Newbold, J. D., J. W. Elwood, R. V. O’Neil & W. Van Winkle, 1982. Nutrient spiralling in streams: Implications for nutrient limitation and invertebrate activity. *Am. Nat.* 120: 628–652.
- Ofenböck, T., O. Moog, J. Gerritsen & M. Barbour, 2004. A stressor specific multimetric approach for monitoring running waters in Austria using benthic macro-invertebrates. *Hydrobiologia* 516: 251–268.
- ÖNORM M 6232, 1997. Richtlinien für die ökologische Untersuchung und Bewertung von Fließgewässern. Österreichisches Normungsinstitut, Wien, 84 pp.
- Rawer-Jost, C., J. Böhmer, J. Blank & H. Rahman, 2000. Macroinvertebrate functional feeding group methods in ecological assessment. *Hydrobiologia* 422/423: 225–232.
- Resh, V. H. & E. P. McElravy, 1993. Contemporary quantitative approaches to biomonitoring using benthic macroinvertebrates. In Rosenberg D. M. & V. H. Resh (eds), *Freshwater Biomonitoring and Benthic Macroinvertebrates*. Chapman and Hall, New York: 159–194.
- Resh, V. H., R. H. Norris & M. T. Barbour, 1995. Design and implementation of rapid assessment approaches for water resource monitoring using benthic macroinvertebrates. *Aust. J. Ecol.* 20: 108–121.
- Resh, V. H., D. M. Rosenberg, & T. B. Reynoldson, 2000. Selection of benthic macroinvertebrate metrics for monitoring water quality of the Fraser River, British Columbia: implications for both multimetric approaches and multivariate models. In Wright J. F., D. W. Sutcliffe & M. T. Furse (eds), *Assessing the Biological Quality of Fresh Waters*. Freshwater Biological Association, Ambleside: 195–206.
- Rolauffs, P., I. Stubauer, S. Zahrádková, K. Brabec & O. Moog, 2004. Integration of the saprobic system into the European Union Water Framework Directive: Case studies in Austria, Germany and Czech Republic. *Hydrobiologia* 516: 285–298.
- Rosenberg, D. M. & V. H. Resh (eds), 1993. Introduction. In Rosenberg, D. M. & V. H. Resh (eds), *Freshwater Biomonitoring and Benthic Macroinvertebrates*. Chapman and Hall, New York: 1–9.
- Rotschein, J., 1982. Nove metody stanovení indexu saprobity. [New methods of saprobic index assessment]. *Vodni hospodárství B* 6: 159–162.
- Roy, A. H., A. D. Rosemond, M. J. Paul & J. B. Wallace, 2003. Stream macroinvertebrate response to catchment urbanisation (Georgia, U.S.A.). *Freshwat. Biol.* 48: 329–346.
- Schweder, H., 1990. Neue Indizes für die Bewertung des ökologischen Zustandes von Fließgewässern abgeleitet aus der Makroinvertebraten-Erährungstypologie. In Fridrich, G. & J. Lacombe (eds), *Ökologische Bewertung von Fließgewässern*. G. Fisher Verlag, Stuttgart, *Limnologie aktuell* 3: 353–377.
- Scarsbrook, M. R. & C. R. Townsend, 1993. Stream community structure in relation to spatial and temporal variation: a habitat template study of two contrasting New Zealand streams. *Freshwat. Biol.* 29: 395–410.
- Schmidt-Kloiber, A. & R. C. Nijboer, 2004. The effect of taxonomic resolution on the assessment of ecological water quality classes. *Hydrobiologia* 516: 269–283.
- Sladeczek, V., 1973. System of water quality from the biological point of view. *Arch. Hydrobiol. Beih. Ergebnisse der Limnologie* 7, pp. 218.
- Sneath, P. H. A. & R. R. Sokal, 1973. *Numerical Taxonomy: the Principles and Practice of Numerical Classification*. Freeman, San Francisco.
- Statzner, B. & F. Sperling, 1993. Potential contribution of system-specific knowledge (SSK) to stream management decision: ecological and economic aspects. *Freshwat. Biol.* 29: 313–342.

- ter Braak, C. J. F., 1986. Canonical correspondence analysis: a new eigenvector technique for multivariate direct gradient analysis. *Ecology* 67: 1167–1179.
- ter Braak, C. J. F. & I. C. Prentice, 1988. A theory of gradient analysis. *Adv. Ecol. Res.* 18: 272–317.
- Verdonschot, P. F. M. & R. C. Nijboer, 2004. Testing the European stream typology of the Water Framework Directive for macroinvertebrates. *Hydrobiologia* 516: 35–54.
- Vinson, M. R. & C. P. Hawkins, 1998. Biodiversity of stream insects: Variation at local, basin and regional scales. *Annu. Rev. Entomol.* 43: 271–293.
- Ward, J. V., 1992. *Aquatic Insect Ecology, Biology and Habitat*. Wiley & Sons, New York: 438 pp.
- Wood, P. J. & P. D. Armitage, 1997. Biological effects of fine sediment in the lotic environment. *Environ. Manag.* 21: 203–217.
- Zelinka, M. & P. Marvan, 1961. Zur Präzisierung der biologischen Klassifikation der Reinheit fließender Gewässer. *Arch. Hydrobiol.* 57: 389–407.